SIMULATED CLINICAL LABORATORY REPORT

TYPE 2 DIABETES MELLITUS - EDUCATIONAL TEST CASE

FICTIONAL / NOT VALID FOR CLINICAL USE. All patient details and measured values are synthetic. The disease, units, and clinical interpretation reflect established medical concepts.

Case information

Case identifier	ABC-TEST-2026-001	Collection date	30 Oct 2026
Patient label	ABC (placeholder)	Age / sex	52 years / adult male (fictional)
Specimen	Venous blood + spot urine	Clinical setting	Outpatient assessment

Reported symptoms and context (synthetic)

Increased thirst (polydipsia), frequent urination (polyuria), fatigue, intermittent blurred vision, and gradual onset over approximately 4 months. Family history of type 2 diabetes; sedentary lifestyle. No acute confusion, vomiting, or rapid breathing described.

Laboratory findings

Test	Result	Reference / diagnostic context	Flag
Fasting plasma glucose	168 mg/dL	Normal 70-99; diabetes >=126 mg/dL	HIGH
HbA1c	8.2 %	Normal <5.7%; diabetes >=6.5%	HIGH
Random plasma glucose	224 mg/dL	>=200 with classic symptoms supports diagnosis	HIGH
Serum creatinine	1.02 mg/dL	Approx. 0.7-1.3 mg/dL (adult male)	Within
eGFR (CKD-EPI 2021)	88 mL/min/1.73 m2	>=90 often normal; interpret in context	Slightly low
Urine albumin/creatinine ratio	42 mg/g	<30 mg/g	HIGH
Total cholesterol	218 mg/dL	Desirable <200 mg/dL	HIGH
LDL cholesterol	142 mg/dL	Optimal <100 mg/dL	HIGH
HDL cholesterol	38 mg/dL	<40 mg/dL is low (men)	LOW
Triglycerides	190 mg/dL	Normal <150 mg/dL	HIGH

Reference intervals vary by laboratory, age, sex, and method. The values above are illustrative, not actual test results.

Clinical interpretation (educational example)

Primary condition: Findings are consistent with type 2 diabetes mellitus with hyperglycemia. Fasting glucose of 168 mg/dL and HbA1c of 8.2% each meet standard diabetes diagnostic thresholds. In an asymptomatic patient, diagnosis generally requires confirmation; classic symptoms with random plasma glucose >=200 mg/dL can establish diagnosis.

Associated findings: Urine albumin/creatinine ratio of 42 mg/g falls within moderately increased albuminuria (A2, 30-300 mg/g); persistence should be confirmed because a single result does not establish chronic kidney disease. Dyslipidemia is also present (elevated LDL and triglycerides, low HDL). An eGFR of 88 alone does not establish kidney disease.

Typical follow-up considerations (not a prescription)

Clinical review of symptoms, blood pressure, weight, medications, and cardiovascular risk; individualized diabetes management plan; repeat urine albumin/creatinine testing to establish persistence; periodic HbA1c and renal function monitoring; foot and dilated eye screening as indicated. Treatment choices require a licensed clinician.

Diagnostic reference concepts

Diabetes thresholds: HbA1c >=6.5%; fasting plasma glucose >=126 mg/dL; 2-hour 75-g OGTT >=200 mg/dL; or random plasma glucose >=200 mg/dL with classic hyperglycemic symptoms. These criteria are from standard ADA diagnostic guidance. Albuminuria categories: A1 <30, A2 30-300, A3 >300 mg/g (KDIGO).

Test-system validation notes

Expected extraction targets: disease name, symptoms, 10 lab measurements, numeric values, units, flags, reference ranges, and follow-up recommendations. No real patient identity, clinician signature, hospital identity, or authentication is present.

Sources for clinical concepts: American Diabetes Association, Standards of Care in Diabetes (Diagnosis and Classification); KDIGO CKD Guideline (albuminuria categories). This document is synthetic and is not an issued medical report.